

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL 1, BL2, BL3)</i>		
Student Name:	T.Muthu Vaishnavi	Reg. No.:	192311215
Section / Batch:	Computer science	Date:	25-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I T.Muthu Vaisnavi (192311215) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T.Muthu Vaisnavi

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Application: Online Video Streaming Platform

A video streaming platform processes user data using the recurrence:

$$T(n) = 2T(n/2) + n$$

We apply the Master Theorem to determine the time complexity.

Master Theorem:

$$T(n) = aT(n/b) + f(n)$$

Here,

$$a = 2$$

$$b = 2$$

$$f(n) = n$$

Step 1: Calculate $n^{\log_b a}$

$$n^{\log_2 2} = n^1 = n$$

Step 2: Compare $f(n)$ with $n^{\log_b a}$

$$f(n) = n$$

$$n^{\log_b a} = n$$

Therefore, $f(n) = \Theta(n^{\log_b a})$.

This matches Case 2 of the Master Theorem.

According to Case 2:

$$T(n) = \Theta(n^{\log_b a} \log n)$$

Therefore,

$$T(n) = \Theta(n \log n)$$

Time Complexity:

The time complexity of the algorithm is $\Theta(n \log n)$.

Scalability:

The algorithm is reasonably scalable as the number of users increases. The $n \log n$ growth rate is more efficient than quadratic time, $\Theta(n^2)$, but less efficient than linear time, $\Theta(n)$. As the number of users increases, the processing time increases at a slightly faster rate than the number of users. Therefore, the platform can handle large amounts of user data efficiently, but performance optimization and distributed processing may be needed for extremely large user populations.

Asymptotic Interpretation:

$\Theta(n \log n)$ gives a tight bound on the algorithm's growth rate. It means that the running time grows proportionally to $n \log n$ for large input sizes, ignoring constants and

lower-order terms. Thus, the algorithm is efficient for processing large user datasets compared with algorithms having quadratic or exponential complexity.

Conclusion:

Using the Master Theorem, the recurrence $T(n) = 2T(n/2) + n$ falls under Case 2, giving a time complexity of $\Theta(n \log n)$. This indicates good scalability for an online video streaming platform as the number of users increases.